

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458328.776 +/- 0.0029 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.079 +/- 0.041
Transit depth (Rp/Rs)^2	0.62 +/- 0.65 [%]
Orbital Inclination (inc)	79.24 +/- 2.82
Airmass coefficient 1 (a1)	0.21 +/- 0.11
Airmass coefficient 2 (a2)	1.47 +/- 0.44
Scatter in the residuals of the lightcurve fit is	47.01 %
Best Comparison Star	None
Optimal Aperture	2.81
Optimal Annulus	3.33
Transit Duration (day)	0.057 +/- 0.041

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R fit vs published	0.079 ± 0.041 vs 0.10538 (deviation 0.49σ)
Duration fit vs published	0.057 d vs 0.1167 d (deviation 1.1σ)
Mid-time O-C	-5.2 min
Depth SNR	1.5
Residual scatter	47.01 %

Light curve

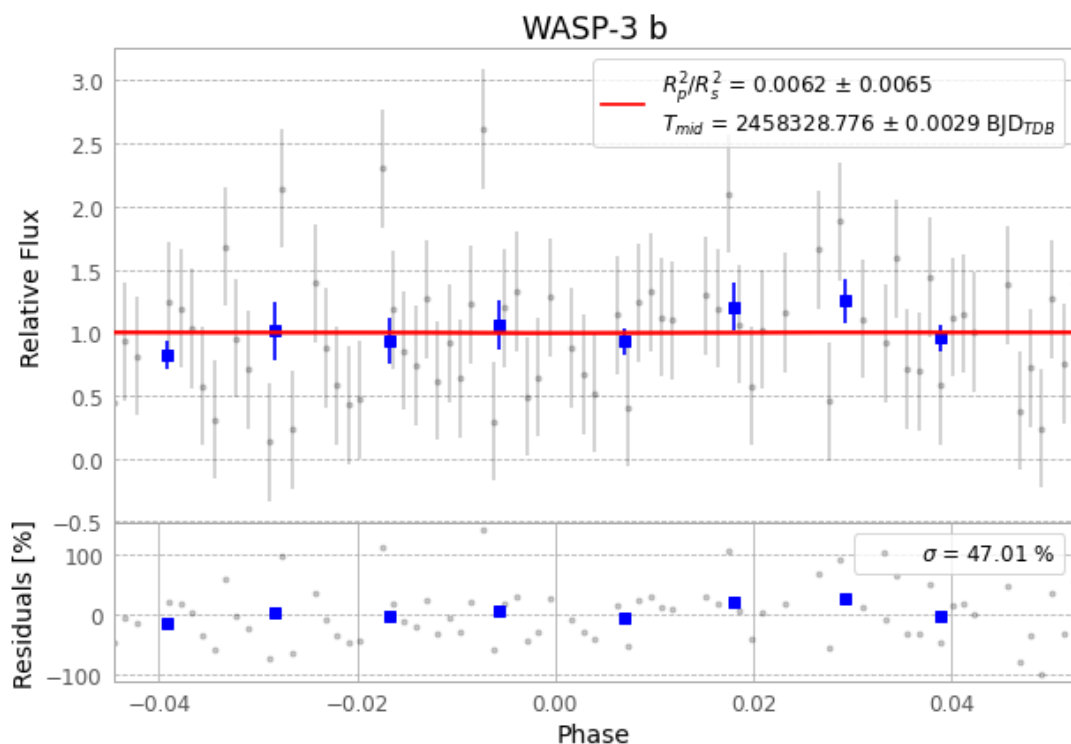


Figure 1 — FinalLightCurve_WASP-3 b_2018-07-28.png (SHA-256 e8e8f438118d83b3...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [372, 127], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 bcb80d35cc5d973e...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 055cf2e

Data provenance

85 FITS frames (56 MB), WASP-3180729043312.FITS → WASP-3180729085112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-3-180729.log (83f24564ac45...), FinalLightCurve_WASP-3 b_2018-07-28.png (e8e8f438118d...), FinalLightCurve_WASP-3 b_2018-07-28.csv (dd890b98eb08...)

Compute resources

Wall time 240.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-17 20:56Z.